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Cambridge, MA (open to hybrid/remote)

EXECUTIVE SUMMARY

Technology leader and computer architect with 30 years across datacenter GPU pathfinding, AI infrastructure research, processor design, and foundry design enablement. Uniquely positioned at the intersection of **3D integration, memory architecture, and the semiconductor supply chain**: pioneered 3D-IC architecture, design and test in academia (a landmark 3D-stacked many-core processor in ISSCC 2012), owned PDK and advanced-packaging design collateral (CoWoS, SoIC, WoW) by leading a three-country design-enablement organization at TSMC, co-developed a near-memory processing system with Samsung at Meta (FAIR), and currently lead memory, interconnect, and co-packaged-optics pathfinding for datacenter AI GPUs at Intel. Proven mentor of industry technical leaders: former students serve as SVP and VP at Samsung, CTO at Intel Government Technology and Lunit, and founders/CEOs of Furiosa AI and Bluecore. Completed an MIT Sloan EMBA to extend deep technical judgment into business and partnership leadership.

Target: a senior leadership/executive role driving 3D-IC and 3D die-stacked memory architecture for next-generation GPU/XPU products, and the memory-vendor, foundry, and OSAT partnerships to deliver innovative solutions for emerging AI applications.

SELECTED AREAS OF EXPERTISE

3D integration: die stacking, TSV, hybrid bonding, 3D DRAM/SRAM, thermal-aware 3D floorplanning, pre-/post-bond test & BISR.
Memory systems: High bandwidth memory and near/in-memory processing, bandwidth/energy co-optimization.
Advanced packaging & design ecosystem: PDK/EDA design methodology, foundry–fabless–memory-vendor partnerships.
Datacenter AI: GPU/XPU scale-up/scale-out architecture, co-packaged optics (CPO) and optical I/O, AI workload characterization, recommendation systems, confidential AI.
Leadership: global R&D organizations (3 countries), research groups built from zero, computer architecture community leadership.

PROFESSIONAL EXPERIENCE

Intel Corporation — Hudson, MA
Intel Fellow, AI System Architecture, Data Center Group — Aug 2022 – Present

- Lead GPU and AI infrastructure technology pathfinding and product incubation for Intel's GPU roadmap.
- Evaluate emerging memory and interconnect technologies for datacenter GPU architecture, spanning HBM/3D-stacked memory, scale-up fabrics, and scale-out networking.
- Led a year-long evaluation of **co-packaged optics (CPO) for GPU scale-up fabrics** with Celestial AI and Ayar Labs, assessing photonic-fabric and optical-I/O chiplet integration, bandwidth density, and energy per bit for future datacenter GPUs.
- Lead AI workload characterization and analysis that informs corporate-level GPU architecture strategy.
- Led confidential AI community building and software stack strategy.
- Led dynamic auto-tuning optimization frameworks for Xeon CPU.

Meta / Facebook AI Research (FAIR) — Cambridge, MA
Area Research Lead (2020–22); Research Head, AI Infrastructure Research (2019–20) — Jan 2019 – Aug 2022

- Founded the "Systems for Machine Learning" research group at FAIR (Boston) from inception and built it into a globally recognized R&D team spanning memory, privacy, and sustainability.
- Led the near-memory processing (NMP) prototype development for at-scale recommendation with Samsung — RecNMP (ISCA'20) and its hardware realization on AxDIMM (IEEE Micro'22), an early industry demo of NMP for DLRM workloads.
- Led secure and privacy-preserving ML system research: SecNDP secure near-data processing (HPCA'22), Cheetah HE acceleration (HPCA'21), MPC private-inference characterization (ISPASS'22), and federated-learning attack analysis (ICML'23).
- Characterized the architecture of Meta's DNN recommendation inference (HPCA'20) and co-led the DeepRecSys / Hercules / RecPipe line of inference-serving systems (ISCA'20, MICRO'21, HPCA'22).
- Initiated and built the program on sustainable, carbon-aware computing for ML: Chasing Carbon (HPCA'21) and ACT carbon-modeling tool (ISCA'22), both selected as **IEEE Micro Top Picks**.

Taiwan Semiconductor Manufacturing Company (TSMC) — Hsinchu, Taiwan
Deputy Director, Design Methodology and Kits Development Division — Apr 2012 – Jan 2019

- Directed **6 departments and 155 R&D staff across Taiwan, China, and the US** developing IC design methodology, EDA tool features, and PDKs for all TSMC customers on all process nodes.
- **Technology owner for all advanced-packaging design collateral (CoWoS, SoIC, and WoW)**: the 2.5D/3D platforms that underpin today's AI accelerators and HBM integration.
- Led design ecosystem enablement across TSMC, EDA vendors, and leading fabless customers (*e.g.,* Apple, NVIDIA, AMD, Broadcom, Qualcomm), negotiating methodology and collateral requirements at the executive level.

Georgia Institute of Technology Atlanta, GA
Associate Professor with tenure (2008–14); Assistant Professor (2002–08), School of ECE Jul 2002 – Jun 2014

- Led **3D-MAPS, the world's first 3D die-stacked many-core processor with stacked memory** (ISSCC 2012; IEEE Trans. Computers 2015), with 3D-stacked memory architecture (HPCA-16), 3D DRAM row cache (TVLSI), and Smart Refresh for 3D DRAM (MICRO-40).
- Established 3D-IC design-for-test: pre-bond testable scan-island architecture (ITC 2007, **ITC 10-Year Most Significant Paper Award**), 3D test wrappers, low-power pre-bond clock trees (ICCAD Best Paper nominee), and 3D memory BISR.
- Pioneered emerging-memory architecture: Security Refresh for PCM wear-out (ISCA-37, **IEEE Micro Top Pick**), SAFER stuck-at-fault recovery (MICRO-43), tri-level-cell PCM (ISCA-40).
- Secured $5.8M in personal federal and industry funding (NSF CAREER, DoE CAREER); built a research group whose alumni now lead at Samsung, Intel, and SanDisk, and have founded two unicorn AI/ML companies (see Mentorship below).

Agere Systems (StarCore DSP Technology Center) Atlanta, GA
Architecture Manager Jul 2001 – Aug 2002

- Managed 10 DSP architects from Agere and Motorola; led StarCore DSP architecture development for 3G infrastructure.

Intel Corporation Santa Clara / Folsom, CA
Senior Processor Architect; Researcher, Microprocessor Research Labs Oct 1995 – Jul 2001

- Processor architect for Pentium III and Timna (integrated CPU/GPU); IA-64/EPIC architecture pathfinding; Pentium III post-silicon validation.
- Authored the cache-optimization chapter of the Intel Architecture Optimization Reference Manual; optimized Microsoft Direct3D 6.1 with SSE.

EDUCATION

Massachusetts Institute of Technology, Sloan School of Management — Master of Business Administration, 2026
University of Michigan, Ann Arbor — Ph.D. (2001) and M.S.E. (1994), Computer Science and Engineering. Horace H. Rackham Distinguished Dissertation Award; nominated for ACM Doctoral Dissertation Award.
National Tsing Hua University, Hsinchu, Taiwan — B.S., Electrical Engineering, 1990 (Class Valedictorian).

MENTORSHIP & TALENT DEVELOPMENT

A consistent record of identifying and developing talent who go on to found companies and run large technical organizations. Selected students I supervised at Georgia Tech include:

- **Joshua B. Fryman, Ph.D.** — CTO & Fellow, Intel Government Technology, Intel, OR, USA.
- **Dong Hyuk Woo, Ph.D.** — Senior Vice President of AGI, Samsung Electronics, CA, USA.
- **Nak Hee Seong, Ph.D.** — Vice President & Head of Exynos Processor Development, Samsung Electronics, South Korea.
- **Chinnakrishnan Ballapuram, Ph.D.** — Senior Director & HBF Base Die Chief Architect, SanDisk, CA, USA.
- **Richard M. Yoo, M.S.** — CTO, Lunit (medical AI), South Korea.
- **Fayez Mohamood, M.S.** — Founder & CEO, Bluecore (unicorn retail-AI company), NY, USA.
- **June Paik, B.S./M.S.** — Founder & CEO, Furiosa AI (unicorn AI accelerator company), South Korea/USA.

HONORS & INDUSTRY LEADERSHIP

- **IEEE Fellow** "for contributions to 3D-ICs and computer architecture," 2017.
- **Distinguished Alumni Award:** EECS College, National Tsing Hua University, Taiwan, 2026.
- **140+ peer-reviewed publications; 45 issued US patents** in memory subsystems, microarchitecture, 3D-IC, physical design, semiconductor devices, machine learning, privacy-enhancing technology, and sustainability. Three IEEE Micro Top Picks; MICRO-33, ANCS-11, and CASES-04 Best Paper Awards.
- **Editor-in-Chief, IEEE Micro (2024–present)**.
- Program Chair, MICRO 2016 and ISCA Industry Track 2021; General Chair, IISWC 2010.
- Inducted into the MICRO and HPCA Halls of Fame.
- Keynotes & Plenary Talks: ISLPED 2019, ISSCC 2016 Design Forum, NSF PIM Workshop 2021, CASPA 2019, ICEIC 2022.

SELECTED SCHOLARLY WORK IN 3D-IC AND MEMORY

3D-MAPS: 3D massively parallel processor with stacked memory (ISSCC 2012; IEEE TC 2015) • Optimized 3D-stacked memory exploiting high-density TSV bandwidth (HPCA 2010) • Scan-island pre-bond testability for die-stacked processors (ITC 2007; 10-Year Most Significant Paper Award) • Test strategies for 3D ICs (IEEE Design & Test 2009) • Near-memory processing in action: AxDIMM (IEEE Micro 2022) • RecNMP (ISCA 2020) • SecNDP: secure near-data processing with untrusted memory (HPCA 2022) • Security Refresh for PCM (ISCA 2010; Top Pick) • Smart Refresh for DRAM (MICRO 2007) • POD: 3D-integrated acceleration layer (IEEE Micro 2008) • At-scale Recommendation Inference (ISCA 2020) • Heterogeneous Inference Serving (HPCA 2022) • Efficient Inference for MoE (NeurIPS 2024)
Complete publication list follows.

COMPLETE PUBLICATION LIST

Selected results are summarized in the sections above. The list below is organized by area; my name is in bold. Journal, conference, and workshop papers are numbered within each area.

Book Chapters

BK1. Sungkap Yeo and **Hsien-Hsin S. Lee**. "Peeling the Power Onion of Data Centers." Chapter Three in *Energy Efficient Thermal Management of Data Centers*, Yogendra Joshi and Pramod Kumar (Eds.), pp. 137–168, Springer, 2012.

BK2. Abderrahim Benquassmi, Eric Fontaine, and **Hsien-Hsin S. Lee**. "Parallelization of Katsevich CT Image Reconstruction Algorithm on Generic Multi-Core Processors and GPGPU." In *GPU Computing GEMS*, Section 10 Medical Imaging, Chapter 31, Wen-Mei Hwu (Editor-in-Chief), pp. 659–677, Morgan Kaufmann Publishers, 2011.

BK3. *Intel Architecture Software Optimization Reference Manual*, Intel Literature Center, order number 245127-001, August 1998. Original author of Chapter 6, "Optimizing Cache Utilization for Pentium III Processor," pp. 6-1 to 6-30, and Appendix A, "The Mathematics of Prefetch Scheduling Distance," pp. F-1 to F-12.

Machine Learning Systems

ML1. Wenjie Xiong, Liu Ke, Maxim Ostapenko, Yongmin Tai, Yeongon Cho, Joonho Song, Jinin So, Kyungsoo Kim, Yongsuk Kwon, Jin Jung, Jieun Lee, Byeongho Kim, Shin-haeng Kang, Sukhan Lee, Jeonghyeon Cho, Kyomin Sohn, Xuan Zhang, **Hsien-Hsin S. Lee**, G. Edward Suh. "Accelerating Confidential Recommendation Model Inference with Near-Memory Processing." *IEEE Transactions on Dependable and Secure Computing*, Vol. 22, Issue 4, pp. 3580–3586, 2025.

ML2. Haiyang Huang, Newsha Ardalani, Anna Sun, Liu Ke, Shruti Bhosale, **Hsien-Hsin S. Lee**, Carole-Jean Wu, Benjamin C. Lee. "Toward Efficient Inference for Mixtures of Experts." 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024), Vancouver, Canada, December 2024.

ML3. Trishita Tiwari, Suchin Gururangan, Chuan Guo, Weizhe Hua, Sanjay Kariyappa, Udit Gupta, Wenjie Xiong, Kiwan Maeng, **Hsien-Hsin S. Lee**, G. Edward Suh. "Information Flow Control in Machine Learning Through Modular Model Architecture." 33rd USENIX Security Symposium, Philadelphia, PA, August 2024.

ML4. Maximilian Lam, Jeff Johnson, Wenjie Xiong, Kiwan Maeng, Udit Gupta, Yang Li, Liangzhen Lai, Ilias Leontiadis, Minsoo Rhu, **Hsien-Hsin S. Lee**, Vijay Janapa Reddi, Gu-Yeon Wei, David Brooks, G. Edward Suh. "GPU-based Private Information Retrieval for On-Device Machine Learning." ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2024), San Diego, CA, April 2024.

ML5. Vivek Parmer, Syed Shakib Sarwar, Ziyun Li, **Hsien-Hsin S. Lee**, Barbara De Salvo, Manan Suri. "Exploring Memory-Oriented Design Optimization of Edge-AI Hardware for Extended Reality Applications." *IEEE Micro*, Special Issue on TinyML, Vol. 43, Issue 6, 2023.

ML6. Sanjay Kariyappa, Chuan Guo, Kiwan Maeng, Wenjie Xiong, Edward Suh, Moinuddin K. Qureshi, and **Hsien-Hsin S. Lee**. "Cocktail Party Attack: Breaking Aggregation-based Privacy in Federated Learning Using Independent Component Analysis." 40th International Conference on Machine Learning (ICML 2023), Honolulu, HI, July 2023.

ML7. Udit Gupta, Mariam Elgamal, Gage Hills, Gu-Yeon Wei, **Hsien-Hsin S. Lee**, David Brooks, Carole-Jean Wu. "Architectural CO2 Footprint Tool: Designing Sustainable Computer Systems With an Architectural Carbon Modeling Tool." *IEEE Micro*, Special Issue on Top Picks from the Computer Architecture Conferences of 2022, Vol. 43, Issue 4, pp. 107–117, 2023.

ML8. Vivek Parmer, Syed Shakib Sarwar, Ziyun Li, **Hsien-Hsin S. Lee**, Barbara De Salvo, Manan Suri. "Memory-Oriented Design Space Exploration of Edge-AI Hardware for XR Applications." tinyML Research Symposium (tinyML 2023), 2023.

ML9. Jiaxun Cui, Xiaomeng Yang, Mulong Luo, Geunbae Lee, Peter Stone, **Hsien-Hsin S. Lee**, Benjamin Lee, G. Edward Suh, Wenjie Xiong, Yuandong Tian. "MACTA: A Multi-agent Reinforcement Learning Approach for Cache Timing Attacks and Detection." 11th International Conference on Learning Representations (ICLR 2023), Kigali, Rwanda, May 2023.

ML10. Mulong Luo, Wenjie Xiong, Geunbae Lee, Yueying Li, Xiaomeng Yang, Amy Zhang, Yuandong Tian, **Hsien-Hsin S. Lee**, G. Edward Suh. "AutoCAT: Reinforcement Learning for Automated Exploration of Cache Timing-Channel Attacks." 29th IEEE International Symposium on High-Performance Computer Architecture (HPCA-29), Montreal, QC, Canada, February 2023.

ML11. Hanieh Hashemi, Wenjie Xiong, Liu Ke, Kiwan Maeng, Murali Annavaram, G. Edward Suh, and **Hsien-Hsin S. Lee**. "Private Data Leakage via Exploiting Access Patterns of Sparse Features in Deep Learning-based Recommendation Systems." Trust and Socially Responsible Machine Learning Workshop, co-located with NeurIPS, December 2022.

ML12. Carole-Jean Wu, Ramya Raghavendra, Udit Gupta, Bilge Acun, Newsha Ardalani, Kiwan Maeng, Gloria Chang, Fiona Aga, James Huang, Charles Bai, Michael Gschwind, Anurag Gupta, Myle Ott, Anastasia Melnikov, David Brooks, Geeta Chauhan, Benjamin Lee, **Hsien-Hsin S. Lee**, Bugra Akyildiz, Maximilian Balandat, Joe Spisak, Ravi Jain, Mike Rabbat, Kim Hazelwood. "Sustainable AI: Environmental Implications, Challenges, and Opportunities." Fifth Conference on Machine Learning and Systems (MLSys), August 2022.

ML13. Udit Gupta, Mariam Elgamal, Gage Hills, Gu-Yeon Wei, **Hsien-Hsin S. Lee**, David Brooks, and Carole-Jean Wu. "ACT: Designing Sustainable Computer Systems with an Architectural Carbon Footprint Modeling Tool." 49th ACM/IEEE International Symposium on Computer Architecture (ISCA-49), pp. 784–799, New York, NY, June 2022.

ML14. Yongqin Wang, Edward Suh, Wenjie Xiong, Benjamin Lefaudeux, Brian Knot, Murali Annavaram, and **Hsien-Hsin S. Lee**. "Characterization of MPC-based Private Inferences for Transformer-based Models." IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS 2022), pp. 187–197, May 2022.

ML15. Udit Gupta, Young Geun Kim, Sylvia Lee, Jordan Tse, **Hsien-Hsin S. Lee**, Gu-Yeon Wei, David Brooks, Carole-Jean Wu. "Chasing Carbon: The Elusive Environmental Footprint of Computing." *IEEE Micro*, Top Picks from the 2021 Computer Architecture Conferences, pp. 37–47, July–August 2022.

ML16. Wenjie Xiong, Liu Ke, Dimitrije Jankov, Michael Kounavis, Xiaochen Wang, Eric Northup, Jie Amy Yang, Bilge Acun, Carole-Jean Wu, Ping Tak Peter Tang, G. Edward Suh, Xuan Zhang, and **Hsien-Hsin S. Lee**. "SecNDP: Secure Near-Data Processing with Untrusted Memory." 28th IEEE International Symposium on High-Performance Computer Architecture (HPCA-28), pp. 244–258, Seoul, South Korea, April 2022.

ML17. Liu Ke, Udit Gupta, Mark Hempstead, Carole-Jean Wu, **Hsien-Hsin S. Lee**, and Xuan Zhang. "Hercules: Heterogeneity-aware Inference Serving for At-scale Personalized Recommendation." 28th IEEE International Symposium on High-Performance Computer Architecture (HPCA-28), pp. 141–154, Seoul, South Korea, April 2022.

ML18. Liu Ke, Xuan Zhang, Jinin So, Jong-Geon Lee, Shin-Haeng Kang, Sukhan Lee, Songyi Han, Yeongon Cho, Jin Hyun Kim, Yongsuk Kwon, Kyungsoo Kim, Jin Jung, Ilkwon Yun, Sung Joo Park, Hyunsun Park, Joonho Song, Jeonghyeon Cho, Kyomin Sohn, Nam Sung Kim, and **Hsien-Hsin S. Lee**. "Near-Memory Processing in Action: Accelerating Personalized Recommendation with AxDIMM." *IEEE Micro*, Special Issue on Processing-In-Memory, January/February 2022.

ML19. Udit Gupta, Samuel Hsia, Jeff Zhang, Mark Wilkening, Javin Pombra, **Hsien-Hsin S. Lee**, Gu-Yeon Wei, Carole-Jean Wu, and David Brooks. "RecPipe: Co-designing Models and Hardware to Jointly Optimize Recommendation Quality and Performance." 54th ACM/IEEE International Symposium on Microarchitecture (MICRO-54), pp. 870–884, Athens, Greece, October 2021.

ML20. Brandon Reagen, Woo-Seok Choi, Yeongil Ko, Vincent T. Lee, **Hsien-Hsin S. Lee**, Gu-Yeon Wei, David Brooks. "Cheetah: Optimizing and Accelerating Homomorphic Encryption for Private Inference." IEEE International Symposium on High-Performance Computer Architecture (HPCA-27), Seoul, South Korea, February 2021.

ML21. Udit Gupta, Young Geun Kim, Sylvia Lee, Jordan Tse, **Hsien-Hsin S. Lee**, Gu-Yeon Wei, David Brooks, Carole-Jean Wu. "Chasing Carbon: The Elusive Environmental Footprint of Computing." IEEE International Symposium on High-Performance Computer Architecture (HPCA-27), Seoul, South Korea, February 2021.

ML22. Udit Gupta, Samuel Hsia, Vikram Saraph, Xiaodong Wang, Brandon Reagen, Gu-Yeon Wei, **Hsien-Hsin S. Lee**, David Brooks, and Carole-Jean Wu. "DeepRecSys: A System for Optimizing End-to-End At-Scale Neural Recommendation Inference." 47th ACM/IEEE International Symposium on Computer Architecture (ISCA-47), Valencia, Spain, 2020.

ML23. Liu Ke, Udit Gupta, Carole-Jean Wu, Benjamin Cho, Mark Hempstead, Brandon Reagen, Xuan Zhang, David Brooks, Vikas Chandra, Utku Diril, Amin Firoozshahian, Bill Jia, Kim Hazelwood, **Hsien-Hsin S. Lee**, Meng Li, Bert Maher, Dheevatsa Mudigere, Maxim Naumov, Martin Schatz, Mikhail Smelyanskiy, and Xiaodong Wang. "RecNMP: Accelerating Personalized Recommendation with Near-Memory Processing." 47th ACM/IEEE International Symposium on Computer Architecture (ISCA-47), Valencia, Spain, 2020.

ML24. Udit Gupta, Carole-Jean Wu, Xiaodong Wang, Maxim Naumov, Brandon Reagen, David Brooks, Bradford Cottel, Kim Hazelwood, Mark Hempstead, Bill Jia, **Hsien-Hsin S. Lee**, Andrey Malevich, Dheevatsa Mudigere, Mikhail Smelyanskiy, Liang Xiong, and Xuan Zhang. "The Architectural Implications of Facebook's DNN-based Personalized Recommendation." 26th IEEE International Symposium on High-Performance Computer Architecture (HPCA-26), San Diego, CA, February 2020.

ML25. Lifeng Nai, Yinglong Xia, Ching-Yung Lin, Bo Hong, and **Hsien-Hsin S. Lee**. "Cache-conscious Graph Collaborative Filtering on Multi-socket Multicore Systems." 11th ACM Conference on Computing Frontiers, Article No. 32, Cagliari, Italy, 2014.

ML26. Richard M. Yoo, Han Lee, Kingsum Chow, and **Hsien-Hsin S. Lee**. "Constructing a Non-Linear Model with Neural Networks for Workload Characterization." IEEE International Symposium on Workload Characterization (IISWC 2006), pp. 150–159, San Jose, CA, October 2006.

Datacenter, Cloud, and Multi-core Computing

DC1. Sungkap Yeo, Mohammad M. Hossain, Jen-Cheng Huang, and **Hsien-Hsin S. Lee**. "ATAC: Ambient Temperature-Aware Capping for Power Efficient Datacenters." ACM Symposium on Cloud Computing, pp. 17.1–17.14, Seattle, WA, 2014.

DC2. Mohammad M. Hossain, Jen-Cheng Huang, and **Hsien-Hsin S. Lee**. "Migration Energy-Aware Workload Consolidation in Enterprise Clouds." IEEE International Conference on Cloud Computing Technology and Science, pp. 405–410, December 2012.

DC3. Sungkap Yeo and **Hsien-Hsin S. Lee**. "SimWare: A Holistic Warehouse-scale Computer Simulator." *IEEE Computer*, Special Issue on Modeling and Simulation of Smart and Green Computing Systems, Vol. 35, No. 9, pp. 48–55, 2012.

DC4. Sungkap Yeo and **Hsien-Hsin S. Lee**. "Using Mathematical Modeling in Provisioning a Heterogeneous Cloud Computing Environment." *IEEE Computer*, Vol. 44, No. 8, pp. 55–62, August 2011.

DC5. Mrinmoy Ghosh, Ripal Nathuji, Min Lee, Karsten Schwan, and **Hsien-Hsin S. Lee**. "Symbiotic Scheduling for Shared Caches in Multi-Core Systems Using Memory Footprint Signature." 40th IEEE International Conference on Parallel Processing (ICPP 2011), pp. 11–20, Taipei, Taiwan, September 2011. (Acceptance rate 22%, 81/363)

DC6. Sung Woo Chung, **Hsien-Hsin S. Lee**, and Woo Hyong Lee. "Architecture/OS Support for Embedded Multi-core Systems." *The Computer Journal*, Vol. 53, No. 8, pp. 1134–1135, 2010.

DC7. Dong Hyuk Woo and **Hsien-Hsin S. Lee**. "PROPHET: Goal-Oriented Provisioning for Highly Tunable Multicore Processors in Cloud Computing." *ACM SIGOPS Operating Systems Review*, special issue on the Interaction among the OS, Compilers, and Multicore Processors, Vol. 43, Issue 2, pp. 102–103, April 2009.

DC8. Michael Healy, **Hsien-Hsin S. Lee**, Gabriel H. Loh, and Sung Kyu Lim. "Thermal Optimization in Multigranularity Multi-Core Floorplanning." 14th IEEE/ACM Asia South Pacific Design Automation Conference (ASP-DAC 2009), pp. 43–48, Yokohama, Japan, January 2009.

DC9. Dong Hyuk Woo and **Hsien-Hsin S. Lee**. "Extending Amdahl's Law for Energy-Efficient Computing in the Many-Core Era." *IEEE Computer*, Vol. 41, No. 12, pp. 24–31, December 2008.

DC10. Richard M. Yoo and **Hsien-Hsin S. Lee**. "Adaptive Transaction Scheduling for Transactional Memory Systems." 20th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA), Special Track on Hardware and Software Techniques to Improve the Programmability of Multicore Machines, pp. 169–178, Munich, Germany, 2008.

DC11. Richard M. Yoo, Yang Ni, Adam Welc, Bratin Saha, Ali-Reza Adl-Tabatabai, and **Hsien-Hsin S. Lee**. "Kicking the Tires of Software Transactional Memory: Why the Going Gets Tough." 20th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA), Special Track on Hardware and Software Techniques to Improve the Programmability of Multicore Machines, pp. 265–274, Munich, Germany, 2008.

DC12. Richard M. Yoo and **Hsien-Hsin S. Lee**. "Helper Transactions: Enabling Thread-Level Speculation via A Transactional Memory System." Workshop on Parallel Execution of Sequential Programs on Multi-core Architectures (PESPMA), in conjunction with ISCA-35, Beijing, China, June 2008.

DC13. Taeweon Suh, Daehyun Kim, and **Hsien-Hsin S. Lee**. "Cache Coherence Support for Non-Shared Bus Architecture on Heterogeneous MP SoCs." 42nd Design Automation Conference (DAC-42), pp. 553–558, Anaheim, CA, June 2005. (Acceptance rate 20%)

DC14. Taeweon Suh, **Hsien-Hsin S. Lee**, and Douglas M. Blough. "Integrating Cache Coherence Protocols for Heterogeneous Multiprocessor Systems, Part 2." *IEEE Micro*, Special Issue on Embedded Systems: Architecture, Design and Tools, pp. 70–78, September/October 2004.

DC15. Taeweon Suh, **Hsien-Hsin S. Lee**, and Douglas M. Blough. "Integrating Cache Coherence Protocols for Heterogeneous Multiprocessor Systems, Part 1." *IEEE Micro*, Special Issue on Embedded Systems: Architecture, Design and Tools, pp. 33–41, July/August 2004.

DC16. Taeweon Suh, Douglas M. Blough, and **Hsien-Hsin S. Lee**. "Supporting Cache Coherence in Heterogeneous Multiprocessor Systems." Design Automation and Test in Europe Conference (DATE 2004), pp. 1150–1155, Paris, France, February 2004. (Acceptance rate 23.2%, 181/780)

GPU and Heterogeneous Architecture

GP1. Jen-Cheng Huang, Joo Hwan Lee, Hyesoon Kim, and **Hsien-Hsin S. Lee**. "GPUMech: GPU Performance Modeling Technique Based on Interval Analysis." 47th IEEE/ACM International Symposium on Microarchitecture (MICRO-47), pp. 268–279, Cambridge, England, December 2014.

GP2. Jen-Cheng Huang, Lifeng Nai, Hyesoon Kim, and **Hsien-Hsin S. Lee**. "TBPoint: Reducing Simulation Time for Large Scale GPGPU Kernels." 28th International Parallel and Distributed Processing Symposium (IPDPS), pp. 437–446, Phoenix, AZ, 2014.

GP3. Hong Jun Choi, Dong Oh Son, Seung Gu Kang, Jong Myon Kim, **Hsien-Hsin S. Lee**, and Cheol Hong Kim. "An Efficient Scheduling Scheme Using Estimated Execution Time for Heterogeneous Computing Systems." *Journal of Supercomputing*, 65(2), pp. 886–902, 2013.

GP4. Dong Hyuk Woo and **Hsien-Hsin S. Lee**. "COMPASS: A Programmable Data Prefetcher Using Idle GPU Shaders." 16th IEEE International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS XV), pp. 297–309, Pittsburgh, PA, March 2010. (Acceptance rate 17.7%, 32/181)

GP5. Ahmad Sharif and **Hsien-Hsin S. Lee**. "Total Recall: A Debugging Framework for GPUs." ACM SIGGRAPH/Eurographics Workshop on Graphics Hardware (GH-08), pp. 13–20, Sarajevo, Bosnia-Herzegovina, June 2008.

GP6. Eric Fontaine and **Hsien-Hsin S. Lee**. "Optimizing Katsevich Image Reconstruction Algorithm on Multicore Processors." 13th IEEE International Conference on Parallel and Distributed Systems (ICPADS-07), Hsinchu, Taiwan, December 2007.

GP7. Dong Hyuk Woo, Joshua B. Fryman, Allan D. Knies, Marsha Eng, and **Hsien-Hsin S. Lee**. "POD: A Parallel On-Die Architecture." 11th Annual Workshop on High Performance Embedded Computing (HPEC), Lexington, MA, September 2007. (Award Session)

GP8. Dong Hyuk Woo, Joshua B. Fryman, Allan D. Knies, and **Hsien-Hsin S. Lee**. "Chameleon: Virtualizing Idle Acceleration Cores of A Heterogeneous Multi-Core Processor for Caching and Prefetching." *ACM Transactions on Architecture and Code Optimization*, Vol. 7, No. 1, pp. 1–35, April 2010.

Energy-Efficient Design

EE1. Mrinmoy Ghosh, Simon Ford, Emre Ozer, Stuart Biles, and **Hsien-Hsin S. Lee**. "Way Guard: A Segmented Counting Bloom Filter Approach to Reducing Energy for Set-Associative Caches." International Symposium on Low Power Electronics and Design (ISLPED-09), pp. 165–170, San Francisco, CA, August 2009. (Selected as one of seven ISLPED highlight papers for publicity and press.)

EE2. Hrishikesh Amur, Ripal Nathuji, Mrinmoy Ghosh, Karsten Schwan, and **Hsien-Hsin S. Lee**. "IdlePower: Application-Aware Management of Processor Idle States." Workshop on Managed Many-Core Systems (MMCS), co-located with ACM/IEEE HPDC, Boston, MA, June 2008.

EE3. Chinnakrishnan S. Ballapuram, Ahmad Sharif, and **Hsien-Hsin S. Lee**. "Exploiting Access Semantics and Program Behavior to Reduce Snoop Power in Chip Multiprocessors." 13th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS XIII), pp. 60–69, Seattle, WA, 2008. (Acceptance rate 24.4%, 31/127)

EE4. Chinnakrishnan S. Ballapuram and **Hsien-Hsin S. Lee**. "Improving TLB Energy for Java Applications on JVM." International Symposium on Systems, Architectures, Modeling and Simulation (SAMOS VIII), pp. 218–223, Samos, Greece, 2008.

EE5. Mrinmoy Ghosh and **Hsien-Hsin S. Lee**. "Virtual Exclusion: An Architectural Approach to Reducing Leakage Energy in Caches for Multiprocessor Systems." 13th IEEE International Conference on Parallel and Distributed Systems (ICPADS-07), Hsinchu, Taiwan, December 2007.

EE6. Dong Hyuk Woo, Mrinmoy Ghosh, Emre Ozer, Stuart Biles, and **Hsien-Hsin S. Lee**. "Reducing Energy of Virtual Cache Synonym Lookup using Blooming Filters." International Conference on Compilers, Architecture, Synthesis for Embedded Systems (CASES 2006), pp. 179–189, Seoul, Korea, October 2006. (Regular paper acceptance rate 24.3%, 25/103)

EE7. Chinnakrishnan S. Ballapuram, Kiran Puttaswamy, Gabriel H. Loh, and **Hsien-Hsin S. Lee**. "Entropy-based Low Power Data TLB Design." International Conference on Compilers, Architecture, Synthesis for Embedded Systems (CASES 2006), pp. 304–311,

Seoul, Korea, October 2006. (Short paper acceptance rate 39.8%, 41/103)

EE8. Mrinmoy Ghosh and **Hsien-Hsin S. Lee**. "DRAMdecay: Using Decay Counters to Reduce Energy Consumption in DRAMs." 3rd Watson Conference on Interaction between Architecture, Circuits and Compilers ($P=AC^2$), Yorktown Heights, NY, October 2006.

EE9. Mrinmoy Ghosh, Emre Ozer, Stuart Biles, and **Hsien-Hsin S. Lee**. "Efficient System-on-Chip Energy Management with a Segmented Bloom Filter." Architecture of Computing Systems (ARCS 2005), pp. 283–297, Frankfurt, Germany, March 2006. (Acceptance rate 21%)

EE10. Chinnakrishnan S. Ballapuram, **Hsien-Hsin S. Lee**, and Milos Prvulovic. "Synonymous Address Compaction for Energy Reduction in Data TLB." ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED-05), pp. 357–362, San Diego, CA, August 2005. (Acceptance rate 22%, 53/233)

EE11. Mrinmoy Ghosh, Weidong Shi, and **Hsien-Hsin S. Lee**. "CoolPression — A Hybrid Significance Compression Technique for Reducing Energy in Caches." IEEE International System-On-Chip Conference (SOCC 2004), pp. 399–402, Santa Clara, CA, September 2004.

EE12. Yuvraj S. Dhillon, Abdulkadir U. Diril, Abhijit Chatterjee, and **Hsien-Hsin S. Lee**. "An Algorithm for Achieving Minimum Energy Consumption in CMOS Circuits Using Multiple Supply and Threshold Voltages at the Module Level." International Conference on Computer-Aided Design (ICCAD-03), pp. 693–700, San Jose, CA, November 2003. (Acceptance rate 26%, 130/490)

EE13. Joshua B. Fryman, Chad M. Huneycutt, **Hsien-Hsin S. Lee**, Kenneth M. Mackenzie, and David E. Schimmel. "Energy Efficient Network Memory for Ubiquitous Devices." *IEEE Micro*, special issue on Power Complexity Aware Design, pp. 60–70, September/October 2003.

EE14. **Hsien-Hsin S. Lee**and Chinnakrishnan S. Ballapuram. "Energy Efficient D-TLB and Data Cache using Semantic-Aware Multilateral Partitioning." International Symposium on Low Power Electronics and Design (ISLPED'03), pp. 306–311, Seoul, Korea, August 2003. (Acceptance rate 24.4%, 54/221)

EE15. **Hsien-Hsin S. Lee**, Joshua B. Fryman, A. Utku Diril, and Yuvraj S. Dhillon. "The Elusive Metric for Low-Power Architecture Research." Workshop on Complexity-Effective Design (WCED-03), held in conjunction with the 30th ACM/IEEE International Symposium on Computer Architecture (ISCA-30), San Diego, CA, June 2003. (Acceptance rate 25%)

EE16. **Hsien-Hsin S. Lee**and Gary S. Tyson. "Region-based Caching: an Energy-Delay Efficient Memory Architecture for Embedded Processors." International Conference on Compilers, Architecture, and Synthesis for Embedded Systems (CASES 2000), pp. 120–127, San Jose, CA, November 2000.

Memory Systems

MM1. Nak Hee Seong, Sungkap Yeo, and **Hsien-Hsin S. Lee**. "Tri-Level-Cell Phase Change Memory: Toward an Efficient and Reliable Memory System." 40th International Symposium on Computer Architecture (ISCA-40), pp. 440–451, Tel-Aviv, Israel, June 2013.

MM2. Sungkap Yeo, Nak Hee Seong, and **Hsien-Hsin S. Lee**. "Can Multi-Level Cell PCM Be Reliable and Usable? Analyzing the Impact of Resistance Drift." 10th Annual Workshop on Duplicating, Deconstructing and Debunking, in conjunction with the 39th International Symposium on Computer Architecture, Portland, OR, June 2012.

MM3. Nak Hee Seong, Dong Hyuk Woo, Vijayalakshmi Srinivasan, Jude A. Rivers, and **Hsien-Hsin S. Lee**. "SAFER: Stuck-At-Fault Error Recovery for Memories." 43rd ACM/IEEE International Symposium on Microarchitecture (MICRO-43), pp. 115–124, Atlanta, GA, December 2010. (Acceptance rate 18%, 45/248)

MM4. Ahmad Sharif and **Hsien-Hsin S. Lee**. "Data Prefetching by Exploiting Global and Local Access Patterns." *Journal of Instruction-Level Parallelism*, Special Issue: The First JILP Data Prefetching Championship (DPC-1), Vol. 13, 2011.

MM5. Nak Hee Seong, Dong Hyuk Woo, and **Hsien-Hsin S. Lee**. "Security Refresh: Prevent Malicious Wear-out and Increase Durability for Phase-Change Memory with Dynamically Randomized Address Mapping." *IEEE Micro*, Special Issue on Top Picks from the Computer Architecture Conferences of 2010, pp. 119–127, January/February 2011.

MM6. Nak Hee Seong, Dong Hyuk Woo, and **Hsien-Hsin S. Lee**. "Security Refresh: Prevent Malicious Wear-out and Increase Durability for Phase-Change Memory with Dynamically Randomized Address Mapping." 37th IEEE International Symposium on Computer Architecture (ISCA-37), pp. 383–394, Saint-Malo, France, June 2010. (Acceptance rate 18%, 44/245)

MM7. Ahmad Sharif and **Hsien-Hsin S. Lee**. "Data Prefetching Mechanism by Exploiting Global and Local Access Patterns." First Journal of Instruction-Level Parallelism Data Prefetching Championship (DPC-1), in conjunction with the 15th IEEE International Symposium on High Performance Computer Architecture (HPCA-15), Raleigh, NC, February 2009.

MM8. Mrinmoy Ghosh and **Hsien-Hsin S. Lee**. "Smart Refresh: An Enhanced Memory Controller Design for Reducing Energy in Conventional and 3D Die-Stacked DRAMs." 40th IEEE/ACM International Symposium on Microarchitecture (MICRO-40), pp. 134–145, Chicago, IL, December 2007. (Acceptance rate 21%, 35/166)

MM9. Xiaotong Zhuang and **Hsien-Hsin S. Lee**. "Reducing Cache Pollution via Dynamic Data Prefetch Filtering." *IEEE Transactions on Computers*, Vol. 56, No. 1, pp. 18–31, January 2007.

MM10. Xiaotong Zhuang and **Hsien-Hsin S. Lee**. "A Hardware Based Cache Pollution Filtering Mechanism for Aggressive Prefetches." International Symposium on Parallel Processing (ICPP-03), pp. 286–293, Kaohsiung, Taiwan, October 2003. (Acceptance rate 20% in architecture track)

MM11. **Hsien-Hsin S. Lee**, Gary S. Tyson, and Matthew K. Farrens. "Bandwidth Utilization using Eager Writeback." *Journal of Instruction-Level Parallelism*, Vol. 4, 2001.

MM12. **Hsien-Hsin S. Lee**, Gary S. Tyson, and Matthew K. Farrens. "Eager Writeback — a Technique for Improving Bandwidth Utilization." 33rd ACM/IEEE International Symposium on Microarchitecture (MICRO-33), pp. 11–21, Monterey, CA, December 2000. (**Best Paper Award of MICRO-33**)

3D-IC Design and Test

3D1. Tianjian Li, Yan Han, **Hsien-Hsin S. Lee**, and Li Jiang. "Fault Clustering Technique for 3D Memory BISR." ACM/IEEE Design Automation & Test in Europe (DATE), Lausanne, Switzerland, March 2017.

3D2. Daehyun Kim, Krit Athikulwongse, Michael B. Healy, Mohammad M. Hossain, Moongon Jung, Ilya Khorosh, Gokul Kumar, Young-Joon Lee, Dean L. Lewis, Tzu-Wei Lin, Chang Liu, Shreepad Panth, Mohit Pathak, Minzhen Ren, Guanhao Shen, Taigon Song, Dong Hyuk Woo, Xin Zhao, Joungho Kim, Ho Choi, Gabriel H. Loh, **Hsien-Hsin S. Lee**, Sung Kyu Lim. "Design and Analysis of 3D-MAPS (3D Massively Parallel Processor with Stacked Memory)." *IEEE Transactions on Computers*, 64(1), pp. 112–125, 2015.

3D3. Dong Hyuk Woo, Nak Hee Seong, and **Hsien-Hsin S. Lee**. "Pragmatic Integration of An SRAM Row Cache in Heterogeneous 3D DRAM Architecture using TSV." *IEEE Transactions on Very Large Scale Integrated Circuits and Systems*, Vol. 21, No. 1, pp. 1–13, January 2013.

3D4. Dae Hyun Kim, Krit Athikulwongse, Michael B. Healy, Mohammad M. Hossain, Moongon Jung, Ilya Khorosh, Gokul Kumar, Young-Joon Lee, Dean L. Lewis, Tzu-Wei Lin, Chang Liu, Shreepad Panth, Mohit Pathak, Minzhen Ren, Guanhao Shen, Taigon Song, Dong Hyuk Woo, Xin Zhao, Joungho Kim, Ho Choi, Gabriel H. Loh, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "3D-MAPS: 3D Massively Parallel Processor with Stacked Memory." Technical Digest of the IEEE International Solid-State Circuits Conference (ISSCC), pp. 188–190, San Francisco, CA, 2012.

3D5. Hong Jun Choi, Young Jin Park, **Hsien-Hsin S. Lee**, and Cheol Hong Kim. "Adaptive Dynamic Frequency Scaling for Thermal-Aware 3D Multi-core Processors." 12th International Conference on Computational Science and Its Applications, pp. 602–612, Salvador de Bahia, Brazil, 2012.

3D6. Xiaodong Wang, Dilip Vasudevan, and **Hsien-Hsin S. Lee**. "Global Built-In Self-Repair for 3-D Memories with Redundancy Sharing and Parallel Testing." IEEE International 3D System Integration Conference (3DIC 2012), Osaka, Japan, 2012.

3D7. Dean L. Lewis, Shreepad Panth, Xin Zhao, Sung Kyu Lim, and **Hsien-Hsin S. Lee**. "Designing 3D Test Wrappers for Pre-bond and Post-bond Test of 3D Embedded Cores." XXIX IEEE International Conference on Computer Design (ICCD-11), pp. 90–95, University of Massachusetts, Amherst, October 2011.

3D8. Xin Zhao, Dean Lewis, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "Low-Power Clock Tree Design for Pre-Bond Testing of 3D Stacked ICs." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, Vol. 30, No. 5, pp. 732–745, 2011.

3D9. Dong Hyuk Woo, Nak Hee Seong, and **Hsien-Hsin S. Lee**. "Heterogeneous Die Stacking of SRAM Row Cache and 3-D DRAM: An Empirical Design Evaluation." 54th IEEE International Midwest Symposium on Circuits and Systems, pp. 1–4, Seoul, Korea, August 2011. (Invited paper)

3D10. Dong Hyuk Woo, Nak Hee Seong, Dean L. Lewis, and **Hsien-Hsin S. Lee**. "An Optimized 3D-Stacked Memory Architecture by Exploiting Excessive, High-Density TSV Bandwidth." 16th IEEE International Symposium on High-Performance Computer Architecture (HPCA-16), pp. 429–440, Bangalore, India, January 2010. (Acceptance rate 18%)

3D11. Michael B. Healy, Krit Athikulwongse, Rohan Goel, Mohammad M. Hossain, Dae Hyun Kim, Young-Joon Lee, Dean L. Lewis, Tzu-Wei Lin, Chang Liu, Moongon Jung, Brian Ouellette, Mohit Pathak, Hemant Sane, Guanhao Shen, Dong Hyuk Woo, Xin Zhao, Gabriel H. Loh, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "Design and Analysis of 3D-MAPS: A Many-Core 3D Processor with Stacked Memory." IEEE Custom Integrated Circuits Conference (CICC), San Jose, CA, September 2010.

3D12. Dean Lewis, Michael Healy, Mohammad Hossain, Tzu-Wei Lin, Mohit Pathak, Hemant Sane, Sung Kyu Lim, Gabriel Loh, and **Hsien-Hsin S. Lee**. "Design and Test of 3D-MAPS, a 3D Die-Stack Many-Core Processor." First IEEE International Workshop on Testing Three-Dimensional Stacked Integrated Circuits (poster), Austin, TX, November 2010.

3D13. **Hsien-Hsin S. Lee**and Krishnendu Chakrabarty. "Test Strategies for 3D Integrated Circuits." *IEEE Design & Test of Computers*, Special Issue on 3D IC Design and Test, Vol. 26, No. 5, pp. 26–35, September/October 2009.

3D14. Xin Zhao, Dean L. Lewis, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "Pre-bond Testable Low-Power Clock Tree Design for 3D Stacked ICs." International Conference on Computer-Aided Design (ICCAD-09), pp. 184–190, San Francisco, CA, November 2009. (**Nominated for the Best Paper Award**)

3D15. Dean L. Lewis and **Hsien-Hsin S. Lee**. "Architectural Evaluation of 3D Stacked RRAM Caches." IEEE International 3D Systems Integration Conference (3DIC-09), pp. 1–4, San Francisco, CA, September 2009.

3D16. Dean L. Lewis and **Hsien-Hsin S. Lee**. "Testing Circuit-Partitioned 3D IC Designs." IEEE Computer Society Annual Symposium on VLSI (ISVLSI-09), pp. 139–144, Tampa, FL, May 2009.

3D17. Dean L. Lewis, Sudhakar Yalamanchili, and **Hsien-Hsin S. Lee**. "High Performance Non-blocking Switch Design in 3D Die-Stacking Technology." IEEE Computer Society Annual Symposium on VLSI (ISVLSI-09), pp. 25–30, Tampa, FL, May 2009.

3D18. Dong Hyuk Woo, Joshua B. Fryman, Allan D. Knies, Marsha Eng, and **Hsien-Hsin S. Lee**. "POD: A 3D-integrated Broad-Purpose Acceleration Layer." *IEEE Micro*, Special Issue on Accelerator Architectures, pp. 28–40, July/August 2008.

3D19. Dean L. Lewis and **Hsien-Hsin S. Lee**. "A Scan-Island Based Design Enabling Pre-bond Testability in Die-Stacking Microprocessors." International Test Conference (ITC 2007), pp. 1–8, Santa Clara, CA, October 2007. (**ITC 2017 10-Year Most Significant Paper Award**)

3D20. Michael Healy, Mario Vittes, Mongkol Ekpanyapong, Chinnakrishnan Ballapuram, Sung Kyu Lim, **Hsien-Hsin S. Lee**, and Gabriel H. Loh. "Multi-Objective Microarchitectural Floorplanning For 2D and 3D ICs." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, Vol. 26, No. 1, pp. 38–52, 2007.

Generic Microarchitecture

uA1. Fayez Mohamood, Mrinmoy Ghosh, and **Hsien-Hsin S. Lee**. "DLL-Conscious Instruction Fetch Optimization for SMT Processors." *Journal of Systems Architecture*, 54, pp. 1089–1100, 2008.

uA2. Eric Fontaine and **Hsien-Hsin S. Lee**. "Bicephaly: Maximizing Bandwidth by Duplexing Power and Data." Workshop on Wild and Crazy Ideas (WACI-VI), in conjunction with ASPLOS XIII, Seattle, WA, 2008.

uA3. Fayez Mohamood, Michael Healy, Sung Kyu Lim, and **Hsien-Hsin S. Lee**. "A Floorplan-Aware Dynamic Inductive Noise Controller for Reliable Processor Design." ACM/IEEE International Symposium on Microarchitecture (MICRO-39), pp. 3–14, Orlando, FL, December 2006. (Acceptance rate 24.1%, 42/174)

uA4. Mongkol Ekpanyapong, Jacob Minz, Thaisiri Watewai, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "Profile-Guided Microarchitectural Floorplanning for Deep Submicron Processor Design." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, Vol. 25, No. 7, pp. 1289–1300, July 2006.

uA5. Fayez Mohamood, Mrinmoy Ghosh, and **Hsien-Hsin S. Lee**. "DLL-Conscious Instruction Fetch Optimization for SMT Processors." 2nd Watson Conference on Interaction Between Architecture, Circuits and Compilers (P=AC^2), pp. 143–152, Yorktown Heights, NY, September 2005. (**Best Paper Award of P=AC^2**)

uA6. Mongkol Ekpanyapong, Sung Kyu Lim, Chinnakrishnan Ballapuram, and **Hsien-Hsin S. Lee**. "Wire-driven Microarchitectural Design Space Exploration." IEEE International Symposium on Circuits and Systems (ISCAS-05), pp. 1867–1870, Kobe, Japan, May 2005.

uA7. Mongkol Ekpanyapong, Pinar Korkmaz, and **Hsien-Hsin S. Lee**. "Choice Predictor for Free." 9th IEEE Asia-Pacific Computer Systems Architecture Conference (ACSAC 2004), pp. 399–413, Beijing, China, September 2004.

uA8. Mikhail Smelyanskiy, Scott A. Mahlke, Edward S. Davidson, and **Hsien-Hsin S. Lee**. "Predicate-aware Scheduling: A Technique for Reducing Resource Constraints." First Annual IEEE/ACM International Symposium on Code Generation and Optimization (CGO 2003), pp. 168–177, San Francisco, CA, March 2003. (Acceptance rate 35.4%)

uA9. **Hsien-Hsin S. Lee**, Mikhail Smelyanskiy, Chris J. Newburn, and Gary S. Tyson. "Stack Value File: Custom Microarchitecture for the Stack." 7th IEEE International Symposium on High Performance Computer Architecture (HPCA-7), pp. 5–14, Monterrey, Mexico, January 2001. (Acceptance rate 23.6%, 26/110)

uA10. Paul Zagacki, Deep Buch, Emile Hsieh, Daniel Melaku, Vladimir Pentkovski, and **Hsien-Hsin S. Lee**. "Architecture of a 3D Software Stack for Peak Pentium III Processor Performance." *Intel Technology Journal*, Vol. 3, Issue 2, May 1999.

Physical Design

PD1. **Hsien-Hsin S. Lee**. "IC Design Challenges and Opportunities for Advanced Process Technology." International Symposium on VLSI Design, Automation, and Test (VLSI-DAT 2015), Hsinchu, Taiwan, April 2015.

PD2. Michael B. Healy, Fayez Mohamood, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "Integrated Microarchitectural Floorplanning and Runtime Controller for Inductive Noise Mitigation." *ACM Transactions on Design Automation of Electronic Systems*, Vol. 16, Issue 4, pp. 46:1–25, 2011.

PD3. Michael Healy, Fayez Mohamood, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "A Unified Methodology for Power Supply Noise Reduction in Modern Microarchitecture Design." 13th IEEE/ACM Asia South Pacific Design Automation Conference (ASP-DAC'08), pp. 611–616, Seoul, Korea, 2008.

PD4. Fayez Mohamood, Michael Healy, Sung Kyu Lim, and **Hsien-Hsin S. Lee**. "Noise-Direct: A Technique for Power Supply Noise Aware Floorplanning Using Microarchitecture Profiling." 12th Asia and South Pacific Design Automation Conference (ASP-DAC'07), pp. 786–791, Yokohama, Japan, January 2007. (Acceptance rate 32.1%, 131/408)

PD5. Michael Healy, Mario Vittes, Mongkol Ekpanyapong, Chinnakrishnan Ballapuram, Sung Kyu Lim, **Hsien-Hsin S. Lee**, and Gabriel H. Loh. "Microarchitectural Floorplanning Using Performance and Temperature Tradeoff." Design, Automation and Test in Europe (DATE 2006), pp. 1288–1293, Munich, Germany, March 2006. (Acceptance rate 17%)

PD6. Mongkol Ekpanyapong, Jacob R. Minz, Thaisiri Watewai, **Hsien-Hsin S. Lee**, and Sung Kyu Lim. "Profile-Guided Microarchitectural Floorplanning for Deep Submicron Processor Design." 41st Design Automation Conference (DAC 2004), pp. 634–639, San Diego, CA, June 2004. (Acceptance rate 21%, 163/785)

Security Architecture and Dependable Systems

SA1. Jen-Cheng Huang, Matteo Monchiero, Yoshio Turner, and **Hsien-Hsin S. Lee**. "Ally: OS-Transparent Packet Inspection Using Sequestered Cores." ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS-11), pp. 1–11, Brooklyn, NY, October 2011. (Acceptance rate 32%, 20/62) (**Best Paper Award of ANCS-11**)

SA2. Jun Yang, Lan Gao, Youtao Zhang, Marek Chrobak, and **Hsien-Hsin S. Lee**. "A Low-Cost Memory Remapping Scheme for Address Bus Protection." *Journal of Parallel and Distributed Computing*, Vol. 70, Issue 5, pp. 443–457, May 2010.

SA3. Vikas R. Vasisht and **Hsien-Hsin S. Lee**. "SHARK: Architectural Support for Autonomic Protection Against Stealth by Rootkit Exploits." 41st ACM/IEEE International Symposium on Microarchitecture (MICRO-41), pp. 106–116, Lake Como, Italy, November 2008. (Acceptance rate 19%, 40/210)

SA4. Weidong Shi and **Hsien-Hsin S. Lee**. "Accelerating Memory Decryption and Authentication with Frequent Value Prediction." ACM International Conference on Computing Frontiers (CF'07), pp. 35–46, Ischia, Italy, May 2007.

SA5. Weidong Shi, Chenghuai Lu, and **Hsien-Hsin S. Lee**. "Memory-centric Security Architecture." Invited article, *Transactions on High Performance Embedded Architectures and Compilers*, Vol. 1, pp. 95–115, Springer Verlag, 2007.

SA6. Dong Hyuk Woo and **Hsien-Hsin S. Lee**. "Analyzing Performance Vulnerability due to Resource Denial-of-Service Attack on Chip Multiprocessors." Workshop on Chip Multiprocessor Memory Systems and Interconnects (CMPMSI), in conjunction with HPCA-13, Phoenix, AZ, February 2007.

SA7. Weidong Shi and **Hsien-Hsin S. Lee**. "Authentication Control Point and its Implications for Secure Processor Design." ACM/IEEE International Symposium on Microarchitecture (MICRO-39), pp. 103–112, Orlando, FL, December 2006. (Acceptance rate 24.1%, 42/174)

SA8. Chenghuai Lu, Tao Zhang, Weidong Shi, and **Hsien-Hsin S. Lee**. "M-TREE: A High Efficiency Security Architecture for Protecting Integrity and Privacy of Software." *Journal of Parallel and Distributed Computing*, Special Issue on Security in Grid and Distributed Systems, Vol. 66, Issue 9, pp. 1116–1128, 2006. (Acceptance rate 12.2%, 10/82)

SA9. Weidong Shi, **Hsien-Hsin S. Lee**, Richard M. Yoo, and Alexandra Boldyreva. "A Digital Rights Enabled Graphics Processing System." ACM SIGGRAPH/Eurographics Workshop on Graphics Hardware, pp. 17–26, Vienna, Austria, September 2006. (Acceptance rate 31.1%, 14/45)

SA10. Lan Gao, Jun Yang, Marek Chrobak, Youtao Zhang, San Nguyen, and **Hsien-Hsin S. Lee**. "A Low-cost Memory Remapping Scheme for Address Bus Protection." 15th International Conference on Parallel Architecture and Compilation Techniques (PACT'06), pp. 74–83, Seattle, WA, September 2006. (Acceptance rate 25.6%, 30/117)

SA11. Weidong Shi, **Hsien-Hsin S. Lee**, Laura Falk, and Mrinmoy Ghosh. "An Integrated Framework for Dependable and Revivable Architecture Using Multicore Processors." 33rd International Symposium on Computer Architecture (ISCA-33), pp. 102–113, Boston, MA, June 2006. (Acceptance rate 13%, 31/231)

SA12. Weidong Shi, Joshua B. Fryman, Guofei Gu, **Hsien-Hsin S. Lee**, Youtao Zhang, and Jun Yang. "InfoShield: A Security Architecture for Protecting Information Usage in Memory." 12th International Conference on High Performance Computer Architectures (HPCA-12), pp. 225–234, Austin, TX, February 2006. (Acceptance rate 14%)

SA13. Weidong Shi, **Hsien-Hsin S. Lee**, Chenghuai Lu, and Mrinmoy Ghosh. "Towards the Issues in Architectural Support for Protection of Software Execution." *ACM SIGARCH Computer Architecture News*, Vol. 33, Issue 1, pp. 6–15, March 2005.

SA14. Weidong Shi, Chenghuai Lu, and **Hsien-Hsin S. Lee**. "Memory-centric Security Architecture." International Conference on High Performance Embedded Architectures and Compilers (HiPEAC 2005), pp. 153–168, Barcelona, Spain, November 2005. (Acceptance rate 20.2%, 17/84)

SA15. Weidong Shi, **Hsien-Hsin S. Lee**, Guofei Gu, Mrinmoy Ghosh, Laura Falk, and Trevor Mudge. "Intrusion Tolerant and Self-recoverable Network Service System Using Security Enhanced Chip Multiprocessor." 2nd IEEE International Conference on Autonomic Computing (ICAC-05), pp. 263–273, Seattle, WA, June 2005. (Acceptance rate of regular papers 16.7%, 25/150)

SA16. Weidong Shi, **Hsien-Hsin S. Lee**, Mrinmoy Ghosh, Chenghuai Lu, and Alexandra Boldyreva. "High Efficiency Counter Mode Security Architecture via Prediction and Precomputation." 32nd International Symposium on Computer Architecture (ISCA-32), pp. 14–24, Madison, WI, June 2005. (Acceptance rate 23.2%, 45/194)

SA17. Weidong Shi, **Hsien-Hsin S. Lee**, Chenghuai Lu, and Tao Zhang. "Attacks and Risk Analysis for Hardware Supported Software Copy Protection Systems." 4th ACM Workshop on Digital Rights Management (DRM 2004), pp. 54–62, Washington, D.C., October 2004.

SA18. Weidong Shi, **Hsien-Hsin S. Lee**, Mrinmoy Ghosh, and Chenghuai Lu. "Architectural Support for High Speed Authentication of Shared Memory in Multiprocessor Systems." International Conference on Parallel Architecture and Compilation Techniques (PACT'04), pp. 123–134, Antibes Juan-les-Pins, France, September 2004. (Acceptance rate 18.8%, 23/122)

SA19. Xiaotong Zhuang, Tao Zhang, **Hsien-Hsin S. Lee**, and Santosh Pande. "Hardware Assisted Control Flow Obfuscation for Embedded Processors." International Conference on Compilers, Architecture, Synthesis for Embedded Systems (CASES 2004), pp. 292–302, Washington, D.C., September 2004. (Acceptance rate 25.2%, 31/123) (**Best Paper Award of CASES 2004**)

SA20. Weidong Shi, **Hsien-Hsin S. Lee**, Chenghuai Lu, and Mrinmoy Ghosh. "Toward the Issues in Architectural Support for Protection of Software Execution." Workshop on Architectural Support for Security and Anti-Virus (WASSA), in conjunction with the 11th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS-XI), pp. 1–9, Boston, MA, October 2004.

FPGA

FP1. Taeweon Suh, Shih-Lien L. Lu, and **Hsien-Hsin S. Lee**. "An FPGA Approach to Quantifying Coherence Traffic Efficiency on Multiprocessor Systems." 17th IEEE International Conference on Field Programmable Logic and Applications (FPL 2007), pp. 47–53, Amsterdam, The Netherlands, August 2007. (Acceptance rate 21%) (Nominated for the Best Paper Award)

FP2. Taeweon Suh, **Hsien-Hsin S. Lee**, Shih-Lien Lu, and John Shen. "Coherence Traffic Considered Harmful — An FPGA Approach to Quantifying Coherence Traffic Efficiency on Multiprocessor Systems." International Symposium on Field-Programmable Gate Arrays (FPGA 2007), Monterey, CA, February 2007. (Poster paper)

FP3. Taeweon Suh, **Hsien-Hsin S. Lee**, Shih-Lien Lu, and John Shen. "Initial Observations of Hardware/Software Co-Simulation using FPGA in Architecture Research." Workshop on Architecture Research using FPGA Platforms (WARFP-2006), in conjunction with HPCA-12, Austin, TX, 2006.

FP4. Taeweon Suh, **Hsien-Hsin S. Lee**, Sally A. McKee, and Martin Schulz. "Evaluating System-wide Monitoring Capsule Design Using Xilinx Virtex-II Pro FPGA." Workshop on Architecture Research using FPGA Platforms (WARFP-2005), in conjunction with HPCA-11, San Francisco, CA, 2005.

FP5. Martin Schulz, Brian S. White, Sally A. McKee, **Hsien-Hsin S. Lee**, and Jurgen Jeitner. "Owl: Next Generation System Monitoring." ACM Computing Frontiers 2005, pp. 116–124, Ischia, Italy, May 2005.

FP6. Martin Schulz, Brian White, Sally A. McKee, and **Hsien-Hsin S. Lee**. "A Vision for Next Generation System Monitoring." Workshop on Hardware Performance Monitor Design and Functionality, in conjunction with HPCA-11, San Francisco, CA, 2005.

FP7. Christopher R. Clark, Ripal Nathuji, and **Hsien-Hsin S. Lee**. "Using an FPGA as a Prototyping Platform for Multi-core Processor Applications." Workshop on Architectural Research using FPGA Platforms (WARFP-2005), in conjunction with HPCA-11, San Francisco, CA, 2005.

Performance Evaluation

PE1. Richard M. Yoo, **Hsien-Hsin S. Lee**, Han Lee, and Kingsum Chow. "Hierarchical Means: Single Number Benchmarking with Workload Cluster Analysis." IEEE International Symposium on Workload Characterization (IISWC 2007), pp. 204–213, Boston, MA, September 2007.

PE2. **Hsien-Hsin S. Lee**, Youfeng Wu, and Gary Tyson. "Quantifying Instruction-Level Parallelism Limits on an EPIC Architecture." IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS-2000), pp. 21–27, Austin, TX, April 2000.

PE3. Eric L. Boyd, Waqar Azeem, **Hsien-Hsin S. Lee**, Tien-Pao Shih, Shih-Hao Hung, and Edward S. Davidson. "A Hierarchical Approach to Modeling and Improving the Performance of Scientific Applications on the KSR1." International Conference on Parallel Processing (ICPP-94), pp. 188–192, St. Charles, IL, August 1994.

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EIC1. **Hsien-Hsin S. Lee**. "The CPU Strikes Back." *IEEE Micro*, Vol. 46, No. 4, July/August 2026.

EIC2. **Hsien-Hsin S. Lee**. "The AI-Hardware Flywheel." *IEEE Micro*, Vol. 46, No. 3, May/June 2026.

EIC3. **Hsien-Hsin S. Lee**. "Reframing AI Memory." *IEEE Micro*, Vol. 46, No. 2, March/April 2026.

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